Case Vignettes: Urologic Emergencies

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Block: Endocrine/Urology/Reproduction

Thread: Clinical Medicine

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Further Reading

Manjunath AS, Hofer MD. *Urologic Emergencies*. Med Clin North Am. 2018 Mar;102(2):373-385. doi: 10.1016/j.mcna.2017.10.013. Review.

(**Available online to students via the Scott Library**)

Learning Objectives

By exam time, you should be able to do the following:

1. Define acute urinary retention and describe its presentation, treatment, and possible sequelae.
2. Define obstructive pyelonephritis and describe its presentation, diagnosis, treatment, and possible sequelae.
3. Define priapism and describe its presentation, diagnosis, treatment, and possible sequelae.
4. Define penile fracture and describe its presentation, diagnosis, treatment, and possible sequelae.
5. Define paraphimosis and describe its presentation, diagnosis, treatment, and possible sequelae.
6. Define testicular torsion and describe its presentation, diagnosis, treatment, and possible sequelae.
7. Define Fournier's gangrene and describe its presentation, diagnosis, treatment, and possible sequelae.

Study Questions

1. A patient presents with 6 hours of a persistent and progressively painful erection.
   - What is the likely diagnosis?
   - How do you differentiate between a low-flow and high-flow state?
   - What are potential predisposing factors?
   - How is this treated?
   - What is a possible long-term sequela?

2. A 12-year-old patient is brought into the emergency room with a 2-hour history of an exquisitely tender scrotum.
   - What diagnosis must be first considered?
   - What is the anatomic abnormality associated with this condition?
   - If the physical examination is not diagnostic, what other test can you order?
   - What is the treatment for this patient?
   - What is the critical time period during which this must be corrected?

3. A 17-year-old patient comes into the emergency room with an 8-hour history of right flank pain and fever of 103F. A non-contrasted CT scan demonstrates a 7mm stone in the proximal right ureter with associated hydronephrosis.
   - What laboratories should be ordered?
   - Should you send this patient home to pass the stone? Why or why not?
   - What other intervention(s) must be performed and why?
   - What is the possible outcome if this is not treated expeditiously?

A. Acute Urinary Retention

1. Background
    a. Presents as a sudden, complete inability to void.
    b. Typically associated with considerable suprapubic and lower abdominal discomfort and can cause significant distress for the affected individual.
2. Epidemiology
    a. AUR is thirteen times more common in a person with a penis than without, and the incidence increases with age.
    b. Nearly 10% of people with penises >70 years of age and nearly 33% >80 years of age will develop AUR in a five-year period
3. Etiology
        i. BPH – risk factors include severity of urinary symptoms, increased prostate volume, decreased urine flow rate, and a PSA >2.5.
        ii. Urethral stricture or other anatomical occlusion of the urethra
        iii. Prostate or bladder cancer
        iv. Constipation
        v. Phimosis and paraphimosis
        vi. Urolithiasis
        vii. Pelvic organ prolapse (utero-vaginal prolapse)
        viii. Urethral diverticulum
    a. General – both sexes
        i. Gross hematuria with clot retention
        ii. Neurogenic impairment and detrusor muscle insufficiency.
        iii. Immediately after general or epidural anesthesia
        iv. Following acute spinal cord injury
        v. Incomplete relaxation of the external urethral sphincter during voiding, a phenomenon known as dyssynergia.
        vi. Anticholinergic and sympathomimetic drugs
        vii. Urinary tract infection
4. Diagnosis
    a. Bladder ultrasound or bladder scan
    b. Bladder catheterization – also therapeutic

5. Treatment
    a. Bladder catheterization with irrigation if blood clots present
    b. NB – postobstructive diuresis may develop that monitoring and possible fluid and electrolyte replacement

B. Obstructive Pyelonephritis

1. Background
    a. Defined as a bacterial infection in an obstructed kidney.
    b. Can occur at any level along the ureter and may result from a stone, a tumor, a ureteral stricture, or a congenital obstruction.
    c. *E. coli* is the most common infecting organism, cultured in over 80% of cases. Other likely causative organisms include *Klebsiella, Proteus, Enterobacter, Pseudomonas, and Citrobacter*.
2. Presentation
    a. Many will present with classic symptoms such as renal colic, fever, chills, dysuria, and costovertebral tenderness.
    b. Other patients may present in septic shock, with hypotension and tachycardia.
3. Diagnosis
    a. Basic laboratory evaluation: CBC with differential, BMP, urine analysis, and culture.
    b. More advanced laboratories: Blood gas, lactate
    c. Imaging – look for evidence of renal obstruction and a possible etiology (e.g. stone)
        i. Non contrasted CT scan of the abdomen and pelvis
        ii. Renal ultrasound
4. Treatment
    a. Hospital admission
        i. Invasive monitoring with ICU care if septic
        ii. Fluid and electrolyte replacement
    b. Immediate decompression of the kidney
        i. Ureteral stent
        ii. Percutaneous nephrostomy tube
        iii. Etiology should <u>*not*</u> be addressed during acute episode
    c. Broad spectrum antibiotics, tailored to culture when available

C. Priapism

1. Background
    a. Defined as a persistent penile erection that continues hours beyond, or unrelated to, sexual stimulation and lasting for at least 4 hours.
    b. Incidence of priapism has been estimated to be 1.5 cases per 100,000 people per year.
    c. General etiology: occurs when there is a failure of the engorged corpus cavernosum to drain either due to impaired relaxation or paralysis of the cavernosal smooth muscle or occlusion of the venous outflow.
2. Types
    a. Ischemic Priapism
        i. Definition:
            1. As the corpus cavernosum fills with blood, the increased intracavernosal pressure eventually begins to decrease arterial inflow
            2. Low inflow and low outflow dynamic results in hypoxia, acidosis, and eventually penile compartment syndrome.
        ii. Presentation:
            1. Patients with ischemic priapism present with a painful, prolonged, and fully rigid erection.
            2. Microscopic tissue edema begins to occur at four to six hours while wide structural damage of cavernous smooth muscles occurs after 12 hours.
            3. Irreversible fibrosis of the corpus cavernosum is seen after 48 hours
        iii. Risk factors
            1. Sickle cell trait or disease (most common cause in children)
            2. Malignant tumor (notably leukemia)
            3. Drugs (PDE5i (sildenafil and similar medications), intracavernous injections, alpha blockers, anticoagulants, trazodone, buproprion, cocaine)
            4. Neurologic shock
        iv. Treatment
            1. Corporal aspiration and irrigation with saline
            2. Intracorporal phenylephrine

3. Surgical decompression
4. Untreated ischemic priapism can result in permanent erectile dysfunction and corporal tissue damage.

b. Non-Ischemic Priapism

a. Definition:

1. Consistently high inflow of blood into the corpus cavernosum without reduced outflow
2. Usually is the result of a fistula between the cavernosal artery and corpus cavernosum.

b. Presentation:

1. Patients with non-ischemic priapism present with a partial and non-tender erection.
2. Irreversible necrosis and fibrosis seen in ischemic priapism is not seen in the non-ischemic type
3. Differential between ischemic and non-ischemic priapism can be determined with a corporal blood gas

c. Causes

1. Needle injury resulting in a hole in the cavernosal artery
2. Blunt trauma
3. Congenital arterial malformations
4. Iatrogenic (fistula can be created during penile surgery)

d. Treatment

1. Non-ischemic priapism is not an emergency and 62% of cases will resolve without treatment.
2. Non-ischemic priapism is not generally related to permanent erectile dysfunction.

D. Penile Fracture

1. Background

a. Penile fractures occur when the erect penis is forcibly bent, causing a rupture of the tunica albuginea of the corporal bodies of the penis.

b. Generally occur during sexual intercourse but may be self-inflicted during vigorous masturbation or may occur with other types of blunt trauma to the erect penis.

2. Presentation
    a. Patients will generally describe a distinct feeling as a “pop” when the tunica ruptures, followed by immediate pain, and a sudden loss of the erection.
    b. Usually accompanied by the development of a significant ecchymosis involving the shaft of the penis and substantial penile swelling which has been referred to as an “eggplant deformity.”
3. Diagnosis
    a. Based primarily on the history and physical exam.
    b. Imaging and laboratories have a limited role
4. Treatment
    a. Immediate surgical repair
    b. Possible complications attributable to penile fracture include erectile dysfunction, penile curvature, or the development of penile plaques.

E. Paraphimosis

1. Background
    a. Occurs when the foreskin of an uncircumcised or partially circumcised male becomes retracted behind the coronal sulcus of the glans penis and will not return to its normal position
    b. When the foreskin remains retracted for an extended amount of time, venous and lymphatic outflow is obstructed and the foreskin and glans become increasingly edematous.
    c. With sufficient edema, arterial inflow to the foreskin and to the glans can become occluded.
    d. Over the course of days or even just hours, this can result in local skin necrosis, but may also result in infarction, gangrene, and autoamputation of the glans.
2. Presentation
    a. Presents with penile pain (or in an infant, general irritability), swelling of the foreskin and glans, a constricting band of tissue at the coronal sulcus, and a flaccid shaft.
3. Diagnosis
    a. Based primarily on the history and physical exam.
    b. Imaging and laboratories have a limited role

4. Treatment
    a. Pain control
    b. Manual reduction
    c. Surgical reduction if manual reduction not successful

F. Testicular Torsion

1. Background
    a. Testis torsion is the most common cause of testis loss in the US.
    b. The incidence in males <25 years old is approximately 1:4000.
    c. Torsion more often involves the left testicle.
    d. A contributing factor is believed to be a congenital anomaly called the "bell clapper deformity," which consists of a transverse as opposed to longitudinal lie of the affected testis
    e. The testis salvage rate approaches 100% in patients who undergo detorsion within 6 hours of the start of pain. However there is only a 20% viability rate if detorsion occurs >12 hours; and virtually no viability if detorsion is delayed >24 hours
2. Presentation
    a. Testicular torsion presents with the rapid onset of severe testicular pain and swelling.
    b. The onset of pain may be preceded by trauma, physical activity, or by no activity (e.g. during sleep).
    c. It most often occurs in children or adolescents, but this diagnosis should be considered in evaluating patients with scrotal pain of any age, as it may occasionally occur in people 40-50 years old.
    d. Torsion should be in the differential for any sudden acute scrotal pain or swelling.
3. Diagnosis
    a. Based primarily on the history and physical exam.
    b. The classic physical examination findings are an exquisitely tender testicle with a high, horizontal lie.
    c. Progressive edema and inflammation may ensues, such that after 1224 hours, the entire hemiscrotum appears as a confluent mass without identifiable landmarks.
    d. Scrotal ultrasonography, especially if the diagnosis is questionable, is the single most useful adjunct to the history and physical examination. Doppler

flow may assess arterial flow within the affected testis; if arterial flow is absent, torsion is highly likely.

e. Ultrasonography may also exclude significant testicular trauma, a hernia extending into the scrotum, or acute epididymo-orchitis.

4. Treatment

1. Urgent surgical exploration and orchidopexy, or orchiectomy if the testis is necrotic.

2. The contralateral testis is at risk, and it should undergo an orchidopexy as well in the same surgery.

Fournier's Gangrene

1. Background

a. Fournier's gangrene is a life threatening, necrotizing infection of the male or female perineum.

b. The causative event is usually a mucosal barrier breakdown in the urethra or colon.

c. The majority of cases are seen in older patients, but patients of both sexes with diabetes mellitus are at higher risk for having Fournier's.

d. Other risk factors include obesity, AIDS, malignancy, alcoholism, smoking, and renal failure.

e. The bacterial infection is from mixed facultative and anaerobic organisms including *E. coli, Klebsiella, enterococci, Bacteroides, Fusobacterium,* and *Clostridium*.

2. Presentation

a. Signs and symptoms are usually swift in onset and dramatic in presentation.

b. Patients can present with both systemic and local symptoms.

c. Local symptoms include blisters, bullae, edema, subcutaneous gas, and crepitus. Systemic symptoms include hypotension, fever, tachycardia, and shock.

d. The infection can spread to the anterior abdominal wall, gluteal folds, and the genitals (penis, scrotum, and labia). C. Diagnosis

e. Based primarily on the history and physical exam.

f. CT scan should be ordered to visualize the extent of the infection and to detect for air in the soft tissue indicating a gas-forming organism.

3. Treatment
    a. Includes broad spectrum antibiotics with emergent, extensive surgical debridement and drainage.
    b. Surgical intervention may include cystostomy, colostomy, orchiectomy, and skin grafting. Surgical intervention is based on the depth and extent of the infection as well as the status of the patient.
    c. After surviving the initial infection, many patients will need multiple reconstructive procedures.
    d. Morbidity and mortality from Fournier's gangrene is high.

Answers to Study Questions

These answers are not meant to be all encompassing; rather they are examples of the types of thought processes needed for application of the material.

1. *This patient has a priapism. Possible etiologies include an inherent hypercoagulable abnormality, such as sickle cell disease or trait, or an underlying malignancy; or use of an erectogenic medication. A high-flow versus low-flow state must be differentiated, and if the history is not clear, a corporal blood gas may be helpful. Corporal aspiration and irrigation with saline is the initial treatment, with intracorporal phenylephrine injection administered for persistent painful priapism. Surgical decompression is indicated if nothing else is successful. Long-term irreversible erectile dysfunction may occur for a delay in treatment.*

2. *This patient has an acute scrotum and testicular torsion must be ruled out. This is associated with a bell clapper deformity. If the examination is nondiagnostic, a scrotal ultrasound with Doppler flow should be ordered. Treatment is a scrotal exploration with detorsion and orchidopexy, with the contralateral testis being pexed as well.*

3. *This patient has an infected upper urinary tract with obstruction caused by a stone. She should be admitted to the hospital and started on broad-spectrum antibiotics. Emergent surgical decompression should be planned via a urteral stent or percutaneous nephrostomy tube. If not treated emergently, she may become septic with significant morbidity or mortality resulting.*